روزبه پورجعفریان (۴۰۴۲۰۱۹۸۴)
ریاضی مهندسی پیشرفته
استاد درس: جناب دکتر امامی

تاریخ دریافت: ۱۴۰۴/۰۸/۱۰
تاریخ ارسال: ۱۴۰۴/۰۸/۲۴
تمرین شمارهٔ ۱

1 Solve — ریاضی مهندسی پیشرفته

a. Gauss Elimination

$$\begin{cases} 2x_1+3x_2-x_3+4x_4-2x_5=5 \\ x_1-2x_2+5x_3-3x_4+x_5=4 \\ 3x_1+x_2-4x_3+2x_4-x_5=6 \\ 4x_1-3x_2+2x_3+x_4+5x_5=7 \end{cases}$$

$$\begin{bmatrix} 2 & 3 & -1 & 4 & -2 & 5 \\ 1 & -2 & 5 & -3 & 1 & 4 \\ 3 & 1 & -4 & 2 & -1 & 6 \\ 4 & -3 & 2 & 1 & 5 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 5 & -3 & 1 & 4 \\ 2 & 3 & -1 & 4 & -2 & 5 \\ 3 & 1 & -4 & 2 & -1 & 6 \\ 4 & -3 & 2 & 1 & 5 & 7 \end{bmatrix} \begin{array}{l} R_1 \leftrightarrow R_2 \\ R_2-2R_1 \\ R_3-3R_1 \\ R_4-4R_1 \end{array}$$

$$\rightarrow \begin{bmatrix} 1 & -2 & 5 & -3 & 1 & 4 \\ 0 & 7 & -11 & 10 & -4 & -3 \\ 0 & 7 & -19 & 11 & -4 & -6 \\ 0 & 5 & -18 & 13 & 1 & -9 \end{bmatrix} \begin{array}{l} \\ \\ R_3-R_2 \\ 7R_4-5R_2 \end{array} \rightarrow \begin{bmatrix} 1 & -2 & 5 & -3 & 1 & 4 \\ 0 & 7 & -11 & 10 & -4 & -3 \\ 0 & 0 & -8 & 1 & 0 & -3 \\ 0 & 0 & -71 & 41 & 27 & -43 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1 & -2 & 5 & -3 & 1 & 4 \\ 0 & 7 & -11 & 10 & -4 & -3 \\ 0 & 0 & -8 & 1 & 0 & -3 \\ 0 & 0 & 0 & 257 & 216 & -171 \end{bmatrix}$$

$\rightarrow x_1-2x_2+5x_3-3x_4+x_5=4 \rightarrow x_1=\frac{644}{257}+\frac{56}{257}t$

$\rightarrow 7x_2-11x_3+10x_4-4x_5=-3 \rightarrow x_2=\frac{252}{257}+\frac{43}{257}t$

$\rightarrow -8x_3+x_4=-3 \rightarrow x_3=\frac{75}{257}-\frac{27}{257}t$

$\rightarrow 257x_4+216x_5=-171 \rightarrow x_5=t,\ x_4=\frac{-171-216t}{257}$

---

b. Gauss elimination

$$\begin{cases} x_1-2x_2+3x_3-x_4+2x_5=1 \\ 2x_1-3x_2+x_3-4x_4+x_5=3 \\ 3x_1-5x_2+2x_3-2x_4-3x_5=5 \end{cases}$$

$$\begin{bmatrix} 1 & -2 & 3 & -1 & 2 & 1 \\ 2 & -3 & 1 & -4 & 1 & 3 \\ 3 & -5 & 2 & -2 & -3 & 5 \end{bmatrix} \xrightarrow{R_2-2R_1,\ R_3-3R_1} \begin{bmatrix} 1 & -2 & 3 & -1 & 2 & 1 \\ 0 & 1 & -5 & -2 & -3 & 1 \\ 0 & 1 & -7 & 1 & -9 & 2 \end{bmatrix} \xrightarrow{R_3-R_2} \begin{bmatrix} 1 & -2 & 3 & -1 & 2 & 1 \\ 0 & 1 & -5 & -2 & -3 & 1 \\ 0 & 0 & -2 & 3 & -6 & 1 \end{bmatrix}$$

- $x_4=r,\ x_5=t \Rightarrow -2x_3+3x_4-6x_5=1 \rightarrow -2x_3+3r-6t=1 \rightarrow x_3=(1+6t-3r)/-2$
- $x_2-5x_3-2x_4-3x_5=1 \rightarrow x_2=1+3t+2r-2.5(1+6t-3r)=9.5r-12t-1.5$
- $x_1-2x_2+3x_3-x_4+2x_5=1 \rightarrow x_1=1+2x_2-3x_3+x_4-2x_5 \Rightarrow x_1=15.5r-17t-0.5$

c. $\begin{cases} 2x_1 + x_2 - 3x_3 = 9 \\ x_1 - 2x_2 + 4x_3 = -3 \\ 3x_1 + x_2 + x_3 = 8 \end{cases}$, Cramer method

$$\det \begin{vmatrix} 2 & 1 & -3 \\ 1 & -2 & 4 \\ 3 & 1 & 1 \end{vmatrix} = 2\begin{vmatrix} -2 & 4 \\ 1 & 1 \end{vmatrix} - \begin{vmatrix} 1 & 4 \\ 3 & 1 \end{vmatrix} - 3\begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} = -22$$

$$\frac{\det(x_1)}{\det} = \begin{vmatrix} 9 & 1 & -3 \\ -3 & -2 & 4 \\ 8 & 1 & 1 \end{vmatrix} \frac{-1}{22} = \frac{29}{11}, \quad \frac{\det(x_2)}{\det} = \begin{vmatrix} 2 & 9 & -3 \\ 1 & 3 & 4 \\ 3 & 8 & 1 \end{vmatrix} \frac{-1}{22} = 1, \quad \frac{\det(x_3)}{\det} = \begin{vmatrix} 2 & 1 & 9 \\ 1 & -2 & -3 \\ 3 & 1 & 8 \end{vmatrix} \frac{-1}{22} = \frac{-10}{11}$$

$$\Rightarrow x_1 = \frac{29}{11}, \quad x_2 = 1, \quad x_3 = \frac{-10}{11}$$

2 Inverse $\det = 24 \rightarrow$ inverse ✓

a. $$\begin{bmatrix} 2 & 1 & -1 & 3 \\ 1 & 0 & 2 & -2 \\ 3 & -1 & 1 & 1 \\ 2 & 3 & 0 & 4 \end{bmatrix} \rightarrow \left[\begin{array}{cccc|cccc} 2 & 1 & -1 & 3 & 1 & 0 & 0 & 0 \\ 1 & 0 & 2 & -2 & 0 & 1 & 0 & 0 \\ 3 & -1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 2 & 3 & 0 & 4 & 0 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} R_2 - \frac{R_1}{2} \rightarrow \\ R_3 - \frac{3}{2}R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0.5 & -0.5 & 1.5 & 0.5 & 0 & 0 & 0 \\ 0 & -0.5 & 2.5 & -3.5 & -0.5 & 1 & 0 & 0 \\ 0 & -2.5 & 2.5 & -3.5 & -1.5 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 & -1 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cccc|cccc} 1 & 0.5 & -0.5 & 1.5 & 0.5 & 0 & 0 & 0 \\ 0 & 1 & -5 & 7 & 1 & -2 & 0 & 0 \\ 0 & -5 & 5 & -7 & -3 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & -1 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} R_4 - 2R_2 \\ R_3 + 5R_2 \\ \rightarrow \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0.5 & -0.5 & 1.5 & 0.5 & 0 & 0 & 0 \\ 0 & 1 & -5 & 7 & 1 & -2 & 0 & 0 \\ 0 & 0 & -20 & 28 & 2 & -10 & 2 & 0 \\ 0 & 0 & 11 & -13 & -3 & 4 & 0 & 1 \end{array}\right] \begin{array}{l} \frac{R_3}{20} \rightarrow \\ (R_4 - 11R_3) \div 2.4 \end{array} \left[\begin{array}{cccc|cccc} 1 & 0.5 & -0.5 & 1.5 & 0.5 & 0 & 0 & 0 \\ 0 & 1 & -5 & 7 & 1 & -2 & 0 & 0 \\ 0 & 0 & 1 & -1.4 & -0.1 & 0.5 & -0.1 & 0 \\ 0 & 0 & 0 & 1 & \frac{-19}{24} & \frac{-5}{8} & \frac{11}{24} & \frac{5}{12} \end{array}\right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -0.5 & +0.5 & -0.5 & 0 \\ 0 & 0 & 1 & 0 & \frac{-29}{24} & \frac{-3}{8} & \frac{13}{24} & \frac{7}{12} \\ 0 & 0 & 0 & 1 & \frac{-19}{24} & \frac{-5}{8} & \frac{11}{24} & \frac{5}{12} \end{array}\right] \begin{array}{l} R_1 - 2R_3 \\ +2R_4 \\ \rightarrow \end{array} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{5}{6} & \frac{1}{2} & \frac{-1}{6} & \frac{-1}{3} \\ 0 & 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{-1}{2} & 0 \\ 0 & 0 & 1 & 0 & \frac{-29}{24} & \frac{-3}{8} & \frac{13}{24} & \frac{7}{12} \\ 0 & 0 & 0 & 1 & \frac{-19}{24} & \frac{-5}{8} & \frac{11}{24} & \frac{5}{12} \end{array}\right]$$

A^{-1}

PAPCO

b.
$$\begin{bmatrix}1&2&3\\0&1&4\\2&-1&1\end{bmatrix} \to \left[\begin{array}{ccc|ccc}1&2&3&1&0&0\\0&1&4&0&1&0\\2&-1&1&0&0&1\end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc|ccc}1&2&3&1&0&0\\0&1&4&0&1&0\\0&-5&-5&-2&0&1\end{array}\right] \xrightarrow{R_3+5R_2}$$

$$\left[\begin{array}{ccc|ccc}1&2&3&1&0&0\\0&1&4&0&1&0\\0&0&15&-2&5&1\end{array}\right] \xrightarrow{\frac{R}{15}} \left[\begin{array}{ccc|ccc}1&2&3&1&0&0\\0&1&4&0&1&0\\0&0&1&\frac{-2}{15}&\frac{1}{3}&\frac{1}{15}\end{array}\right] \xrightarrow[R_2-4R_3]{R_1-3R_3} \left[\begin{array}{ccc|ccc}1&2&0&\frac{7}{5}&-1&\frac{-1}{5}\\0&1&0&\frac{8}{15}&\frac{-1}{3}&\frac{-4}{15}\\0&0&1&\frac{-2}{15}&\frac{1}{3}&\frac{1}{15}\end{array}\right]$$

$$\xrightarrow{R_1-2R_2} \left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{3}&\frac{-1}{3}&\frac{1}{3}\\0&1&0&\frac{8}{15}&\frac{-1}{3}&\frac{-4}{15}\\0&0&1&\frac{-2}{15}&\frac{1}{3}&\frac{1}{15}\end{array}\right] \Rightarrow B^{-1}=\begin{bmatrix}\frac{1}{3}&\frac{-1}{3}&\frac{1}{3}\\\frac{8}{15}&\frac{-1}{3}&\frac{-4}{15}\\\frac{2}{15}&\frac{1}{3}&\frac{1}{15}\end{bmatrix}, \quad \det(B)=15$$

3 $A, B \to$ diagonal matrices, order n

a. $AB \to$ diagonal

$$A=\begin{bmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&&\vdots\\\vdots&&\ddots&\vdots\\a_{n1}&\cdots&&a_{nn}\end{bmatrix}_{n\times n} \begin{cases}a_{ij}=0 & i\neq j\\ a_{ij}\neq 0 & i=j\end{cases}, \quad B=\begin{bmatrix}b_{11}&b_{12}&\cdots&b_{1n}\\b_{21}&b_{22}&&\vdots\\\vdots&&\ddots&\vdots\\b_{n1}&\cdots&&b_{nn}\end{bmatrix}_{n\times n} \begin{cases}b_{ij}=0 & i\neq j\\ b_{ij}\neq 0 & i=j\end{cases}$$

Multiply: $C_{ij}=\sum_{k=1}^{n} a_{ik}b_{kj} \to$ column k in row i in A multiplied to row k column j in $B \Rightarrow$

for $i=j \to C_{ii}=\sum a_{ik}b_{ki} \begin{cases}a_{ii}b_{ii} & i=k\\ 0 & i\neq k\end{cases} \Rightarrow C_{ii}=a_{ii}b_{ii} \Rightarrow C=\begin{bmatrix}a_{11}b_{11}&0&\cdots&0\\0&a_{22}b_{22}&&\vdots\\\vdots&&\ddots&\\0&\cdots&&a_{nn}b_{nn}\end{bmatrix}_{n\times n}$

for $i\neq j \to C_{ij}=\sum c_{ik}b_{kj} \to$ zero values for $i\neq j \Rightarrow C_{ij}=0$

b. $AB=BA$

Multiply BA: $D_{ij}=\sum_{k=1}^{n} b_{ik}a_{kj} \to$ column k in row i in B multiplied to row k column j in A.

for $i=j \to D_{ii}=\sum_{k=1} b_{ik}a_{ki} \begin{cases}b_{ii}a_{ii} & i=k\\ 0 & i\neq k\end{cases} \Rightarrow D_{ii}=b_{ii}a_{ii} \Rightarrow D=\begin{bmatrix}b_{11}a_{11}&0&\cdots&0\\0&b_{22}a_{22}&&\vdots\\\vdots&&\ddots&\\0&\cdots&&b_{nn}a_{nn}\end{bmatrix}_{n\times n}$

for $i\neq j \to D_{ij}=\sum_{k=1}^{n} b_{ik}a_{kj} \to$ zero values for $i\neq j \Rightarrow C_{ij}=0$

$\Rightarrow a_{ii}b_{ii}=b_{ii}a_{ii} \Rightarrow C=D \Rightarrow AB=BA$

4 A,B $\Rightarrow$ Squqare order 2 matrix $\Rightarrow adj(AB)=(adj\,B)(adj\,A)$

$(AB)^{-1} = adj(AB)/det(AB)$ if $det(AB)\neq 0 \Rightarrow adj(AB)=(AB)^{-1}det(AB)$

$$\begin{cases}(AB)^{-1}=B^{-1}A^{-1}\\ det(AB)=det(A)det(B)\end{cases} \Rightarrow adj(AB)=B^{-1}A^{-1}det(A)det(B),\ B^{-1}=\frac{adj(B)}{det(B)},\ A^{-1}=\frac{adj(A)}{det(A)}$$

$$\Rightarrow adj(AB)=\frac{adj(B)}{det(B)}\frac{adj(A)}{det(A)}det(A)\,det(B)=adj(B)\,adj(A) \Rightarrow adj(AB)=(adj\,B)(adj\,A)$$

Sample:

$$A=\begin{bmatrix}a & b\\ c & d\end{bmatrix} \Rightarrow adj(A)=\begin{bmatrix}d & -b\\ -c & a\end{bmatrix},\ B=\begin{bmatrix}e & f\\ g & h\end{bmatrix} \Rightarrow adj(B)=\begin{bmatrix}h & -f\\ -g & e\end{bmatrix}$$

$$\Rightarrow adj(B)adj(A)=\begin{bmatrix}hd+fc & -(hb+fa)\\ -(gd+ec) & gb+ea\end{bmatrix}$$

$$AB=\begin{bmatrix}ae+bg & af+bh\\ ce+dg & cf+dh\end{bmatrix} \Rightarrow adj(AB)=\begin{bmatrix}cf+dh & -(af+bh)\\ -(ce+dg) & ae+bg\end{bmatrix}$$

$$\Rightarrow adj(AB)=adj(B)\,adj(A)$$

---

5 nontrivual solution

$$\begin{cases}(5-\lambda)x_1+4x_2+2x_3=0\\ 4x_1+(5-\lambda)x_2+2x_3=0\\ 2x_1+2x_2+(2-\lambda)x_3=0\end{cases}$$

it is form of $(A-I\lambda)x=0 \Rightarrow$ for a nontrivial solution $\Rightarrow det(A-I\lambda)=0$

$$\begin{vmatrix}(5-\lambda) & 4 & 2\\ 4 & (5-\lambda) & 2\\ 2 & 2 & (2-\lambda)\end{vmatrix}=0 \Rightarrow (5-\lambda)\big((5-\lambda)(2-\lambda)-4\big)-4\big((4)(2-\lambda)-4\big)+2\big(8-2(5-\lambda)\big)=0$$

$$\Rightarrow \lambda^3-12\lambda^2+21\lambda-10=0 \Rightarrow \lambda_1=1,\ \lambda_2=1,\ \lambda_3=10$$

$\Rightarrow$ for $\lambda=1,\ \lambda=10$ has nontrivial solution.

6 determinate

$$M(t)=\begin{bmatrix}1 & t & t^2 & t^3\\ 1 & t+1 & (t+1)^2 & (t+1)^3\\ 1 & t+2 & (t+2)^2 & (t+2)^3\\ 1 & t+3 & (t+3)^2 & (t+3)^3\end{bmatrix}\xrightarrow[R_4-R_3]{R_2-R_1,\ R_3-R_2}\begin{bmatrix}1 & t & t^2 & t^3\\ 0 & 1 & (t+1)^2-t^2 & (t+1)^3-t^3\\ 0 & 1 & (t+2)^2-(t+1)^2 & (t+2)^3-(t+1)^3\\ 0 & 1 & (t+3)^2-(t+2)^2 & (t+3)^3-(t+2)^3\end{bmatrix}$$

$(1)\times(\det_{3\times3})$

$$\begin{vmatrix}1 & 2t+1 & 3t^2+3t+1\\ 1 & 2t+3 & 3t^2+9t+7\\ 1 & 2t+5 & 3t^2+15t+19\end{vmatrix}\xrightarrow[R_3-R_2]{R_2-R_1}\begin{vmatrix}1 & 2t+1 & 3t^2+3t+1\\ 0 & 2 & 6t+6\\ 0 & 2 & 6t+12\end{vmatrix}\Rightarrow(1)\begin{vmatrix}2 & 6t+6\\ 2 & 6t+12\end{vmatrix}$$

$\Rightarrow 12t+24-12t-12=12 \Rightarrow \det(M(t))=12$, for no t it is singular.

7

a. if $a_{ij}=r_is_j \Rightarrow A=(a_{ij})$ rank $=1$ or zero

assume $A_{m\times n} \Rightarrow$ assume: $r=[r_1,r_2,\dots,r_m]^T$, $s=[s_1,s_2,\dots,s_n]^T$

$\Rightarrow$ from $a_{ij}=r_is_j$, we have: $\underset{m\times n}{A}=rs^T$

$$\underset{m\times n}{A}=\begin{bmatrix}r_1\\ r_2\\ \vdots\\ r_m\end{bmatrix}_{m\times1}\begin{bmatrix}s_1,s_2,\dots,s_n\end{bmatrix}_{1\times n}=\begin{bmatrix}r_1s_1 & r_1s_2 & \cdots & r_1s_n\\ r_2s_1 & r_2s_2 & \cdots & r_2s_n\\ \vdots & \vdots & \ddots & \\ r_ms_1 & r_ms_2 & & r_ms_n\end{bmatrix}\Rightarrow A_{:,j}=s_jr\quad j=1,\dots,n$$

every column is scalar multiple of the vector r therefor:

$$\operatorname{rank}(A)=\dim(\text{column}(A))\leq 1\quad\begin{cases}\operatorname{rank}(A)=0 & r=0 \text{ or } s=0\\ \operatorname{rank}(A)=1 & \text{otherwise}\end{cases}$$

b. if $A=[a_{ij}]$ is rank 1 $\Rightarrow a_{ij}=r_is_j$

$\operatorname{rank}(A)=1 \Rightarrow$ dimension is 1, one base vector (nonzero) $\Rightarrow r=A_{:,j}$: one of A column nonzero and basis.

$\Rightarrow$ just of basis therefor, other columns are scaled with scaler by s

$\Rightarrow A_{:,j}=s_jr$, if we hae zero column, it's become $s=0$ for that column.

$\Rightarrow s=(s_1,s_2,\dots,s_n)^T \Rightarrow A=rs^T \Rightarrow a_{ij}=r_is_j$, $r=(r_1,\dots,r_m)^T$, $s=(s_1,\dots,s_n)^T$

8 $A \in \mathbb{C}^{n\times n}$, $\text{index}(\lambda)=k$, $\text{alg mult}(\lambda)=m \Rightarrow \dim N((A-\lambda I)^k)=m$

$A = PJP^{-1}$, $J = \text{diag}\left(J_{s_1}(\lambda), J_{s_2}(\lambda), \dots, J_{s_t}\right)$, J_{s_i} : Jordan block with size (s_i)

$\Rightarrow \sum_{i=1}^{t} s_i = m$

for each Jordan block: $J_{s_i}(\lambda) - \lambda I = N_{s_i}$, N_{s_i} : nilpotent matrix.

if $p \geq s_i \Rightarrow N_{s_i}^p = 0 \Rightarrow \dim N(N_{s_i}^p) = s_i$

index of eigenvalues: $\text{index}(\lambda) = k = \max(s_i)$

if $k \geq s_i \Rightarrow N_{s_i}^k = 0 \Rightarrow \dim N((J-\lambda I)^k) = \sum_{i=1}^{t} s_i = m$

$A = PJP^{-1} \Rightarrow (A-\lambda I)^k = P(J-\lambda I)P^{-1} \Rightarrow \dim N((A-\lambda I)^k) = \dim N((J-\lambda I)^k)$

$\Rightarrow$

$\dim N((A-\lambda I)^k) = \dim N((J-\lambda I)^k) = m \Rightarrow \dim N((A-\lambda I)^k) = m$

$\left\{ m = \sum_{i=1}^{t} s_i \geq \max s_i = k \Rightarrow m \geq s_i \Rightarrow N_{s_i}^m = 0 \right\} \Rightarrow \dim N((A-\lambda I)^m) = \sum_{i=1}^{t} s_i = m$

$\Rightarrow \dim N((A-I\lambda)^m) = m$

2

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

a. eigenvalues and eigenvectors.

triangular $\rightarrow (A - I\lambda)x = 0 \rightarrow \det(A - I\lambda) = 0 \rightarrow (2-\lambda)(3-\lambda)(4-\lambda)(5-\lambda) = 0$

$\Rightarrow \lambda_1 = 2,\ \lambda_2 = 3,\ \lambda_3 = 4,\ \lambda_4 = 5$

for $\lambda_1 = 2$: $(A - I\lambda_1)x = 0 \rightarrow$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0 \rightarrow x_1 + x_2 = 0,\ x_3 = x_4 = 0$$

$\Rightarrow v_1 = [1, -1, 0, 0]^T$

$x_1 = 1 \rightarrow x_2 = -1$

for $\lambda_2 = 3$: $(A - I\lambda_2)x = 0 \rightarrow$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0 \rightarrow x_2 = r \Rightarrow x_2 = 1$$

$v_2 = [0, 1, 0, 0]^T$

for $\lambda_3 = 4$: $(A - I\lambda_3)x = 0 \rightarrow$

$$\begin{bmatrix} -2 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0 \rightarrow x_3 = p \Rightarrow x_3 = 1$$

$v_3 = [0, 0, 1, 0]^T$

for $\lambda_4 = 5$: $(A - I\lambda_4)x = 0 \rightarrow$

$$\begin{bmatrix} -3 & 0 & 0 & 0 \\ 1 & -2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0 \rightarrow x_4 = q \rightarrow x_4 = 1$$

$v_4 = [0, 0, 0, 1]^T$

b.

$$P = [v_1 | v_2 | v_3 | v_4] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},\quad D = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix},\quad P^{-1} = \begin{bmatrix} +1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

c. $A = PDP^{-1} \Rightarrow A^5 = ?$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

$$A^5 \Rightarrow D: \text{Diagonal} \Rightarrow D^5 = \begin{bmatrix} 2^5 & 0 & 0 & 0 \\ 0 & 3^5 & 0 & 0 \\ 0 & 0 & 4^5 & 0 \\ 0 & 0 & 0 & 5^5 \end{bmatrix} = \begin{bmatrix} 32 & 0 & 0 & 0 \\ 0 & 243 & 0 & 0 \\ 0 & 0 & 1024 & 0 \\ 0 & 0 & 0 & 3125 \end{bmatrix}$$

$$\Rightarrow A^5 = PD^5P^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 32 & 0 & 0 & 0 \\ 0 & 243 & 0 & 0 \\ 0 & 0 & 1024 & 0 \\ 0 & 0 & 0 & 3125 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 32 & 0 & 0 & 0 \\ 211 & 243 & 0 & 0 \\ 0 & 0 & 1024 & 0 \\ 0 & 0 & 0 & 3125 \end{bmatrix}$$